THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which tissue would be most affected in a plant if the cambium is removed? 1

(a) water absorption **(b)** increase in length
(c) gas exchange **(d)** increase in girth

2. A section of an animal tissue shows flat cells with a single layer. This is likely: 1

(a) bone **(b)** squamous epithelium
(c) columnar epithelium **(d)** cuboidal epithelium

3. A person can bend the arm at the elbow only in one plane. This joint is a: 1

(a) pivot joint **(b)** hinge joint **(c)** ball and socket **(d)** fixed

4. A stem with a girth increasing yearly has active: 1

(a) stomata **(b)** vascular cambium **(c)** root cap **(d)** apical meristem

5. A student sees dead cells with thick lignified walls in a section. The tissue is: 1

(a) phloem **(b)** collenchyma **(c)** parenchyma **(d)** sclerenchyma

6. A tissue of long cells with cilia on the surface is found in the: 1

(a) stomach **(b)** trachea **(c)** kidney **(d)** skin

7. The structure that reduces friction at joints is: 1

(a) cartilage and synovial fluid **(b)** tendon
(c) bone **(d)** muscle

8. Annual rings in a tree trunk are formed in: 1

(a) secondary xylem **(b)** pith **(c)** cork **(d)** phloem

9. A tissue section shows living cells with unevenly thickened corners. The tissue is: 1

(a) collenchyma **(b)** parenchyma **(c)** sclerenchyma **(d)** xylem

10. Cells of tissue in which the matrix is hard due to calcium and phosphorus: 1

(a) blood **(b)** cartilage **(c)** bone **(d)** areolar

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are the cells of the meristematic tissue closely packed with no intercellular spaces? 2

12. Why does cutting the bark in a ring (girdling) kill the tree after some time? 2

13. State two differences between bone and cartilage. 2

14. How do xylem vessels and sieve tubes differ in structure? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how the guard cells control the opening and closing of stomata. 3

16. Explain how the tissue organisation is different in a multicellular plant and an animal with examples of conducting tissues. 3

17. Describe the relationship between tissues, organs and organ systems with the example of the stomach. 3

OR

Differentiate between simple and complex tissues with two examples each. 3

18. Why do leaves have a thick cuticle in desert plants? Which tissue produces it? 3

OR

State the features of cardiac muscles that suit the continuous pumping of the heart. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student cuts a thin section of a young stem and observes it under a microscope. Describe the steps and name the tissues seen from outside to inside, with the function of each. 5

OR

Describe the adaptations of the tissues in aquatic, desert and land plants with examples. 5

20. Describe the structure of the xylem and phloem using a transverse section, the position of the two in a vascular bundle in the root and the stem. 5

OR

Explain how the activity of meristems leads to plant growth in length and girth, and the way grass is able to regenerate after grazing. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND